Highly Available Web Service with Local DNS

Shehab Eldeen Ahmed

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Contents

1 Task Overview

The environment was built as three isolated virtual machines. One machine provides DNS service, while the other two host the same web service. The web servers operate as primary and backup nodes and share a single virtual IP address. Users therefore access one domain name and one service address without needing to know which web server is currently active.

2 Project Structure

The submitted project is organized as follows:

```
linux_task/
├── Vagrantfile
├── provision/
│   ├── dns.sh
│   └── web.sh
├── config/
│   ├── bind/
│   │   ├── db.test.com
│   │   ├── named.conf.local
│   │   └── named.conf.options
│   └── nginx/
│       ├── master.conf
│       ├── backup.conf
│       ├── web1.conf
│       ├── web2.conf
│       ├── web1-index.html
│       └── web2-index.html
├── screenshots/
│   ├── dig_via_the_dns_server.png
│   ├── resloving_the_master.png
│   ├── after_halting_web1.png
│   └── ping.png
└── linux_task_report.tex
```

3 Approach

The environment was created using infrastructure automation so that all three machines could be started and prepared consistently. A dedicated DNS server was used to map the test domain to a virtual IP address. Two web servers were then placed behind that address in a primary–backup arrangement. Health monitoring was used to detect loss of the primary server and transfer the virtual IP address to the backup server automatically.

Validation was carried out in stages: first by checking the virtual machines, then by confirming DNS resolution, accessing the service while the primary was active, and finally stopping the primary server and accessing the same domain again to verify failover.

4 Tools Used

- **Vagrant and VirtualBox** for creating and managing the virtual machines.
- **Ubuntu Linux** as the operating system for all nodes.
- **BIND9** for local DNS resolution.

- **NGINX** for serving the web pages.
- **Keepalived** for health monitoring and virtual IP failover.
- **dig, curl, and ping** for validation and testing.

5 Results

5.1 Virtual Machine Availability

Both web virtual machines were reachable and responsive. This confirmed that the nodes were running successfully before testing the service behavior.

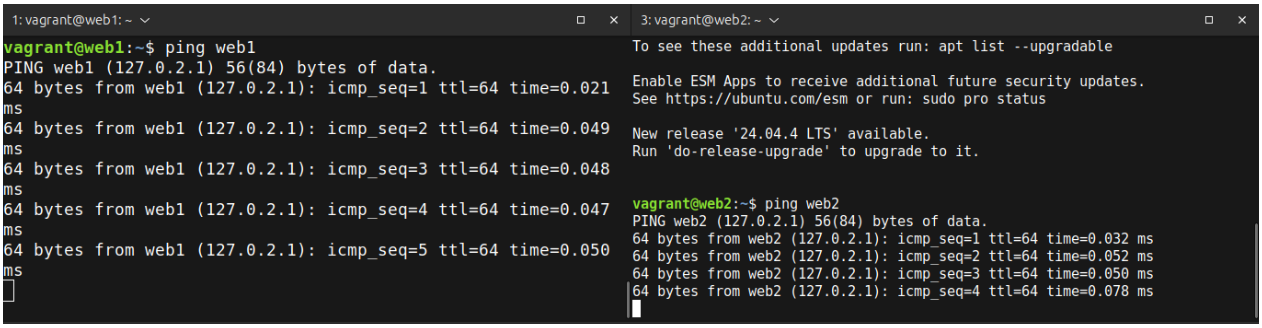

Figure 1: Basic reachability checks on the web-server virtual machines.

5.2 DNS Resolution

The DNS query completed successfully and returned the shared virtual IP address `192.168.56.20` for `test.com`. This shows that clients can locate the service through its domain name rather than a physical web-server address.

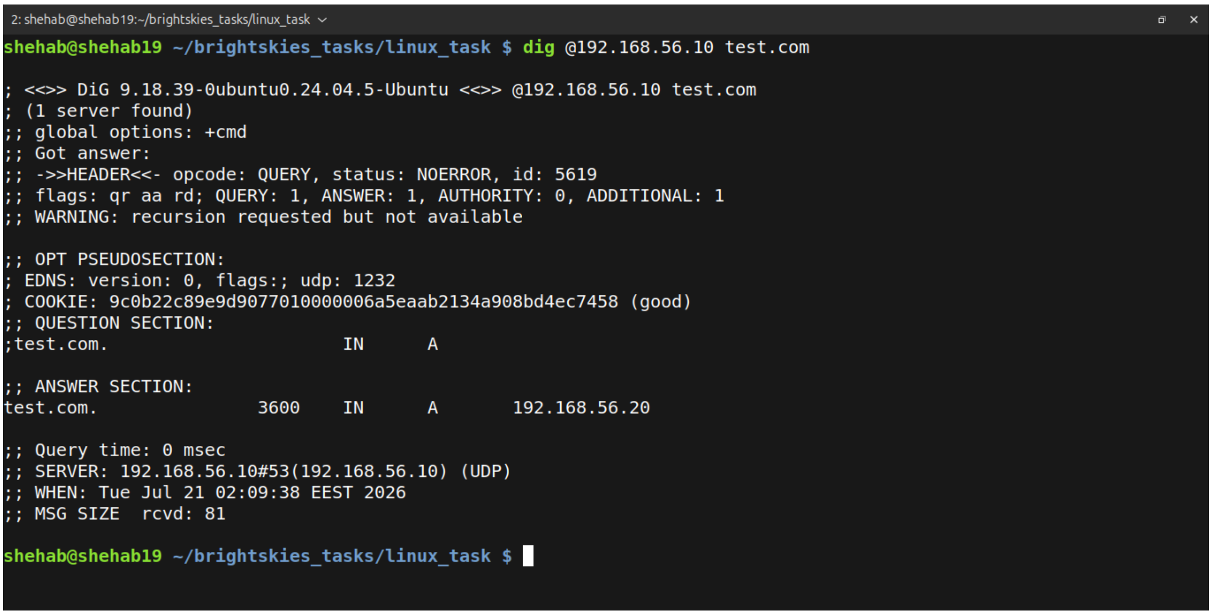

Figure 2: The local DNS server resolves the domain to the virtual IP address.

5.3 Primary Web Server

With both web servers online, a request to the domain reached Web1, confirming that the primary node initially owned the shared service address and served the expected page.

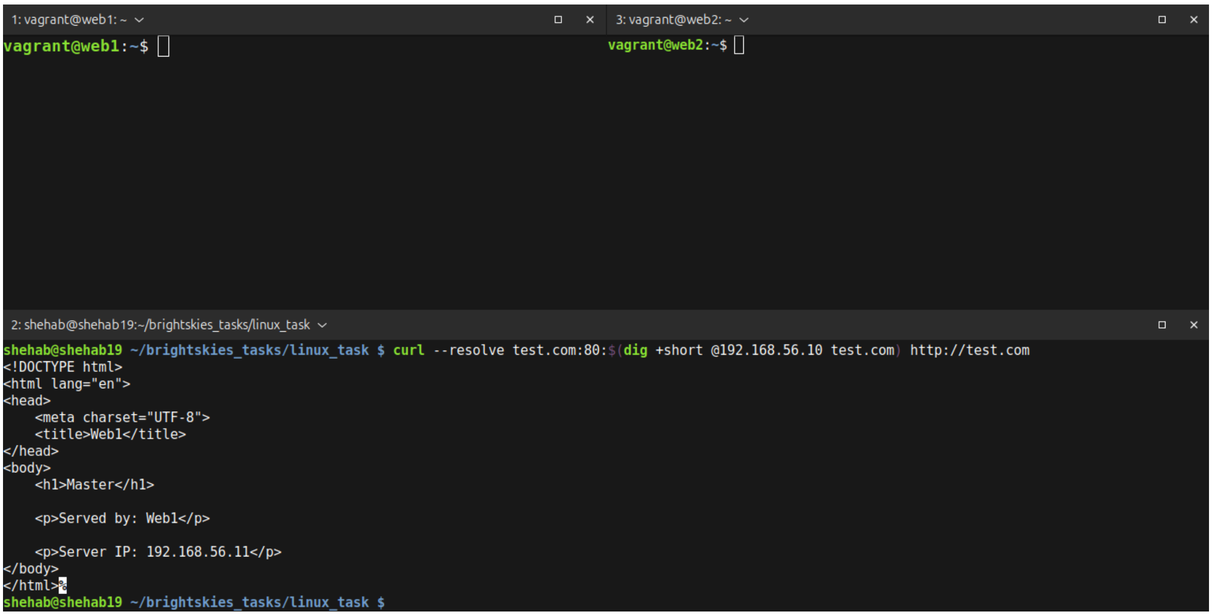

Figure 3: The web service is initially provided by the primary server, Web1.

5.4 Automatic Failover

After Web1 was halted, the same domain and virtual IP address remained available. The response was then served by Web2, demonstrating that the backup node automatically took over the shared address and maintained service availability.

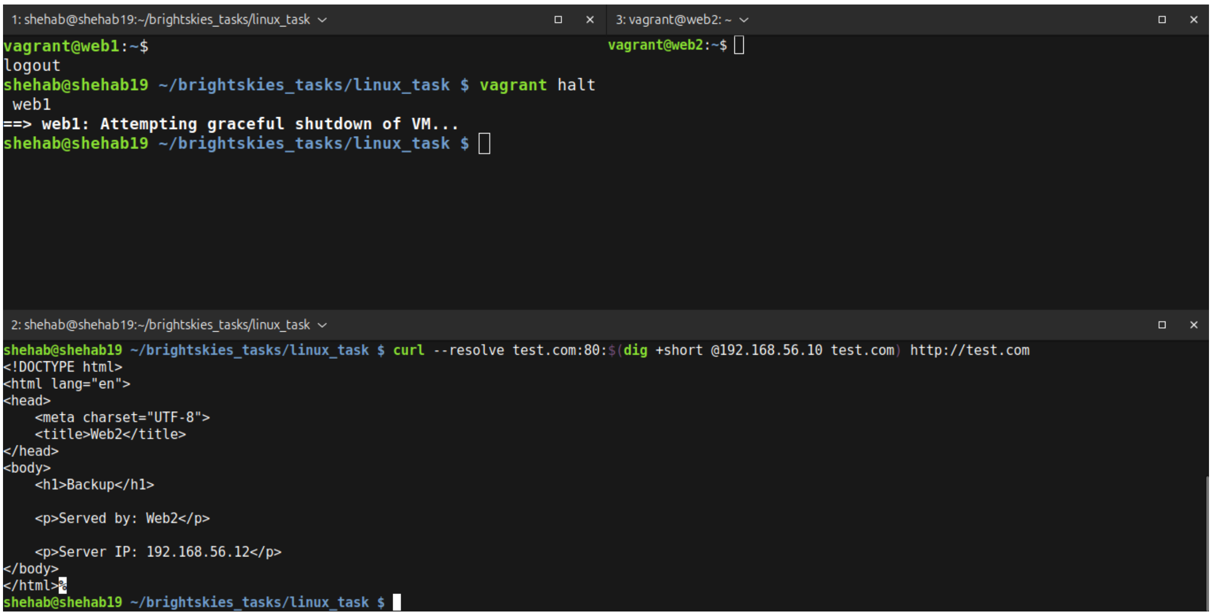

Figure 4: Web2 serves the same domain after the primary server is halted.